

Assignment2

22124187_Zhang Linyihan

Question1

The question uses Weixin API as an example.

1. What the API does?

An API, Application Programming Interface, is a tool that allows different software systems to communicate with each other by following defined rules for requests, data exchange, and responses. The Weixin Official Account API is used to let developers interact with the Weixin public platform through code. According to the documentation, it supports functions such as basic interfaces, message management, user management, material management, custom menu management, account management, data statistics, Weixin Pay, and card-related features.

2. Base URL and key endpoints

The base URL is <https://api.weixin.qq.com/>. Key endpoints shown in the documentation include GET `/cgi-bin/token` for obtaining `access_token`, GET `/cgi-bin/getcallbackip` for getting Weixin server IP addresses, POST `/cgi-bin/callback/check` for network checking, POST `/cgi-bin/clear_quota` for clearing API call quota, POST `/cgi-bin/menu/create` for creating a custom menu, GET `/cgi-bin/menu/get` for querying a custom menu, GET `/cgi-bin/menu/delete` for deleting a custom menu, and POST `/cgi-bin/media/upload` for uploading temporary media.

3. Required parameters and headers

The documentation states that most Weixin API calls require an `access_token` as the credential. For GET requests, parameters are appended directly to the URL. For POST requests, parameters are submitted in JSON format and the header uses `Content-Type: application/json`. For example, the token endpoint requires `grant_type`, `appid`, and `secret`, and the custom menu creation endpoint requires `access_token` in the URL and a JSON request body. The temporary media upload endpoint requires `access_token`, `type`, and a file uploaded with `multipart/form-data`.

4. Sample request/response in JSON

(1) A typical JSON response for getting an access token is:

```
1 {  
2   "access_token": "ACCESS_TOKEN",  
3   "expires_in": 7200
```

```
4 | }
```

(2) A typical JSON request for clear_quota interface is:

```
1 | {  
2 |   "appid": "wx123456"  
3 | }
```

Its sample JSON response is:

```
1 | {  
2 |   "errcode": 0,  
3 |   "errmsg": "ok"  
4 | }
```

5. Explanation of data structure

The Weixin API returns data in JSON format. In the token response, access_token is a string used as the credential for API calls, and expires_in is a number that shows how long the token is valid. In the custom menu request, the outer object contains a button array. Each object in that array represents one top-level menu item. Fields such as type, name, key, url, appid, and pagepath describe the behavior of each menu item, while sub_button is an array used for nested menu items. In general, successful responses contain business data, while failed responses contain errcode and errmsg.

6. Limitations

First, access_token is valid for 7200 seconds, should be cached properly, and repeated retrieval invalidates the previous token. It also recommends using a central server to manage token refresh. Second, the documentation says each public account has daily call limits, and error code 45009 means the API call exceeds the allowed limit. Third, the clear_quota interface can only be called 10 times per month. Fourth, media management has file restrictions: images are limited to 2MB in JPG/PNG format, voice files to 2MB in MP3/AMR/WAV format with playback under 60 seconds, videos to 10MB in MP4 format, thumbnails to 64KB in JPG format, and temporary media is valid for 3 days.

Question2

The question uses ordering food at a restaurant as an example.

1. What the client is

In a RESTful system, the client is the side that sends a request. In the restaurant example, the client is the customer. The customer looks at the menu, chooses what they want, and tells restaurant system their order. In the same way, in a computer system, a client can be a web browser, a mobile app, or another program that asks for information or a service.

2. What the server is

The server is the side that receives the request, processes it, and sends back a result. In the restaurant example, the server is the restaurant staff and kitchen. They receive the order, prepare the food, and return it to the customer. In a RESTful system, the server receives requests from the client, handles them, and returns responses.

3. What a response and request are

A request is the message sent from the client to the server. In the restaurant example, the request is the customer saying, "I would like a burger and a cola." This is like asking the system for something.

A response is the message sent back from the server to the client. In the restaurant example, the response may be "Your order has been accepted," "The burger is unavailable," or the food itself being served. In a RESTful system, the response usually includes the result of the request.

4. What status codes mean

Status codes show the result of a request. They help the client understand whether the request was successful or not. In the restaurant example:

200 OK means the order was accepted successfully.

400 Bad Request means the order was incomplete.

404 Not Found means the requested item is not on the menu.

500 Internal Server Error means something went wrong in the restaurant.

In RESTful systems, these status codes provide a quick and standard way to explain what happened.

5. What JSON is and why it's used

JSON stands for JavaScript Object Notation. It is a simple and structured format used to organize and exchange data. In the restaurant example, JSON shows what the customer ordered. For example:

```
1 {  
2   "food": "burger",  
3   "drink": "cola",  
4   "quantity": 1  
5 }
```

JSON is used because it is easy for both humans and computers to read and understand. In RESTful systems, it is commonly used to send data between the client and the server in a organized way.